

Effi-Ace: Efficient and Accurate Prediction for High-Resolution Spectrum Tenancy

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Abstract—Spectrum prediction is a key enabler for the forthcoming coexistence paradigm where various Radio Access Technologies share overlapping radio spectrum, to substantially improve spectrum efficiency in 5G and beyond systems. Although this fundamental issue has received tremendous research attention, existing algorithms are designed for and validated against spectrum usage data in low time-frequency granularities, which causes inevitable errors when applied to spectrum prediction in realistic resolutions. Therefore, in this paper, we redesign three key components along the spectrum prediction pipeline to propose Effi-Ace, an efficient and accurate prediction for high-resolution spectrum tenancy. First, we obtain raw spectrum data in the same resolutions as scheduling, which reflects the actual dynamics of the subject to be predicted. We improve the Deep Q-Network (DQN) prediction algorithm with enhanced experience replay to reduce sample complexity, so that the proposed DQN is more efficient in terms of sample quantities. New prediction features are extracted from high-resolution measurement data to improve prediction accuracy. According to our detailed experiments, the proposed prediction algorithm substantially reduces sample complexity by 88.9%, and the improvements in prediction accuracy are up to 14%, when compared with various state-of-the-art counterparts.

I. INTRODUCTION

Contrary to the drastically increasing data rates in mobile networks [1], the crowded radio bandwidth shows a lukewarm advancement. Wireless coexistence [2]–[5], is a promising solution to sustain the growth of mobile data rate with limited spectrum [6]–[8]. In the coexistence paradigm, various Radio Access Technologies (RATs) are allowed to work on overlapping frequency channels to improve efficiency. Standardizations have been carried out to guide the development of the coexistence of various RATs, such as IEEE 1900.X series, IEEE 802.22, and IEEE 802.11af. Due to the wide deployment of Long Term Evolution (LTE), the standardization of the coexistence between LTE and WiFi is also developed, such as MulteFire [9] and Licensed Assisted Access (LAA) [10]. As elaborated in the survey [11], a wide range of fundamental operations in the coexistence paradigm, including spectrum mobility prediction, spectrum sensing, and topology control can be tremendously facilitated by spectrum prediction [12]–[15], a technology that infers future spectrum usage [16]–[19].

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Though spectrum prediction has received substantial research efforts, existing algorithms are proposed based on spectrum tenancy in time-frequency granularities much coarser than the scheduling of the corresponding RATs [20], [21]. For example, a prediction algorithm targeting Global System for Mobile Communications (GSM) spectrum usage is trained for and validated against spectrum usage data with a time resolution of 1.8 seconds [22]. However, a GSM scheduling interval is 4.61538 ms, so predicting 1.8 seconds, or 390 actual GSM scheduling intervals of 4.61538 ms as positive by majority voting leads to as much as 195 of those intervals marked as false positives, which *inevitably causes prediction errors due to low resolutions in the raw spectrum usage data*. The low resolution in raw data is even more serious for the spectrum prediction of LTE systems [23], where spectrum resources are dynamically assigned every one ms.

To achieve raw spectrum usage in the same resolution as scheduling, we adopt the decoding based measurement targeting LTE [24]. A Software Defined Radio (SDR) testbed is set up to decode the Downlink Control Information (DCI) of a commercial LTE cell, and the spectrum assignment for all users in the cell is parsed. In this way, we achieve the spectrum tenancy data in the same time resolution as they are assigned. LTE spectrum usage is chosen as the target because it is currently the most widely deployed mobile system, and its spectrum structures and assignment process will stay mostly unchanged in 5G systems. Thus, the prediction based on our data applies to a wide range of systems for years to come.

In addition to the tenancy data in their true resolutions, we for the first time further consider more control channel statistics than history tenancy as prediction features, such as the number of Orthogonal Frequency-Division Multiplexing (OFDM) symbols, the amount of DCI, Resource Assignment (RA) types, and Modulation and Coding Schemes (MCS).

The high resolution spectrum tenancy is volatile due to the capricious nature of mobile networks where sudden changes may result from radio channel conditions, user mobility, and mobile traffic. Thus, we base our prediction algorithm on the Rainbow DQN that adapts its network weights in runtime [25]. As mobile devices are reasonable places to deploy the spectrum prediction algorithm, the sample complexity of the DQN must be small so that the DQN can be trained efficiently

after accumulating a limited amount of experience. Hence, we propose a new experience replay scheme that jointly considers the Temporal Difference (TD) errors and tenancy distances among data samples. In this way, the learning agent is always provided with the samples that are most relevant and not learned well enough. Compared with the state-of-the-art Rainbow DQN, our new experience replay scheme substantially lowers the sample complexity by 88.9%.

In addition to the high resolution spectrum tenancy, we for the first time explore new prediction features enabled by the decoding method. While parsing the LTE DCIs to obtain spectrum assignments, protocol fields closely related to the wireless channel conditions are also decoded, including the number of OFDM symbols, the amount of DCIs, RA types, and the MCS. We conduct feature selections based on the Fisher Score (FS), and identify that the number of DCIs is the best prediction input for spectrum tenancy among the new data types. Since introducing new prediction features could enlarge the size of the state space of the prediction input and thus the sample complexity, we derive an upper bound for the sample complexity of binary prediction based on Markov Decision Process (MDP). We show both analytically and experimentally that the new prediction features improve the prediction accuracy without increasing the sample complexity.

Our main contributions are recapitulated as follows.

- We collect high-resolution spectrum tenancy and related protocol fields in the same time granularity with resource assignment in an LTE cell using the decoding method. The feature selection step identifies the powerful prediction inputs, i.e. spectrum tenancy and the number of DCIs in the previous 40 milliseconds (Sec. II).
- We formulate spectrum tenancy prediction as an MDP (Sec. III), and propose to solve it by an upgraded DQN with a new prioritized experience replay method to consider both the TD errors and the distance between different spectrum tenancy. Thus, the sample complexity is substantially reduced by 88.9% when compared with the state-of-the-art Rainbow DQN, making our algorithm highly sample efficient for mobile deployment with a short ramp up time as low as 20 seconds (Sec. V).
- We derive an upper bound for the sample complexity of MDP based binary prediction (Sec. VI). We show both analytically and experimentally that the new prediction features enhance the prediction accuracy without increasing the sample complexity, which is the first attempt to use other spectrum parameters as prediction inputs than the previous spectrum usage. Compared with various prediction algorithms, such as Rainbow DQN and Multi-Layer Perceptron (MLP) [26], the new features achieve an accuracy improvement as much as 14% (Sec. VII).

II. MEASUREMENT DATA AND FEATURE SELECTION

To achieve high resolution prediction inputs, we carry out the decoding based measurement of a commercial LTE cell, and select prediction features from various measurement data related to spectrum tenancy. Our detailed spectrum tenancy

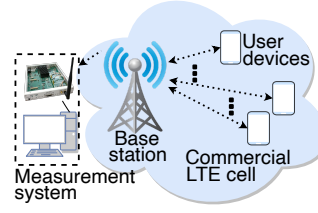


Fig. 1: Measurement setup

TABLE I: Measurement related to spectrum tenancy.

Data type	Range
Binary usage	$\{0, 1\}$
OFDM count	$\{1, 2, 3\}$
DCI count	$\{1, 2, \dots\}$
RA type	$\{0, 1, 2\}$
MCS	$\{0, \dots, 31\}$

data ensures that the prediction model is trained and validated by accurate usage at the same resolutions with scheduling.

A. Raw measurement data

Different from the widely adopted energy detection method, we set up an SDR based measurement system targeting commercial LTE systems to decode the downlink control channels, which is illustrated in Fig. 1. As the LTE base station assigns spectrum usage and airs the decision to mobile users on the control channel, we decode and parse the relevant data using the tool proposed in [24].

Besides the spectrum tenancy that has long been adopted as the sole prediction feature, the decoding based measurement enables many other potential prediction features, such as scheduling information. Among them, we choose to study four quantities, the number of OFDM symbols, the number of DCIs, the RA types, and MCS index, because they are closely related to the assignment of LTE Resource Blocks (RBs). Specifically, the spectrum resources of an LTE cell are assigned every millisecond, and the assignment is contained in the control channel at the beginning of every millisecond. The number of OFDM symbols for the downlink control channel can be one, two, or three. DCIs are inside the control channel region and they contain spectrum resource assignments for mobile users in the cell. In each DCI, the RB assignments in 180 kHz are encoded in three RA types according to LTE standards, i.e., type 0, type 1, and type 2 [27]. DCIs also specify how the data carried on spectrum resources are modulated and coded, i.e., MCS. The MCS values are six-bit long, taking values in the range of $[0, 31]$. The measurement data are also summarized in Table I.

We record five types of spectrum tenancy related data from a commercial LTE cell. The system bandwidth is 10 MHz that accommodates 50 LTE RBs. We accumulate the data over four days that have an equal amount of workdays and rest days. Thus, we end up with 17280 million samples of tenancy, MCS, and RA types for which one data point is required per RB. Since the number of OFDM symbols of the control channel and the number of DCIs are assigned once per millisecond for all RBs, we obtain 345.6 million samples. The experiments in this paper are based on this data set unless stated otherwise. Compared with other spectrum predictions [22], [26], our data set measures spectrum usage the scheduling resolution, and more diverse data types are recorded as potential candidates for feature selections than the binary tenancy.

B. Feature selection

Due to the fine time granularity and diverse data types in our raw measurement results, adopting the samples in several previous minutes as the prediction features, as most other prediction work do [22], [26], may introduce many candidates with low prediction capabilities. Thus, we conduct feature selections to choose the best prediction inputs.

We employ FS as the selection metrics to evaluate the five types of measurement data in Table I. FS algorithm rewards features that assign similar values to samples corresponding to the same output and disparate values to samples with different output classes [28]. Thus, for a classification with M classes, the FS of a feature F_i is

$$F_i = \frac{\sum_{j=1}^M n_j (\mu_{ij} - \mu_i)^2}{\sum_{j=1}^M n_j \rho_{ij}^2}, \quad (1)$$

where n_j is the number of the instances of feature F_i corresponding to class j . The mean value of all instances of feature F_i is μ_i , and the mean of F_i s in class j is μ_{ij} . The variance of F_i s in class j is ρ_{ij} . Large FS values indicate high prediction capabilities of the features.

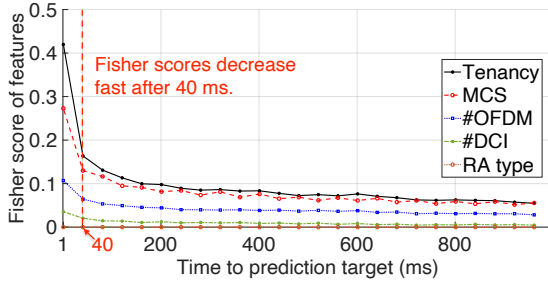


Fig. 2: Fisher Score of five types of prediction features.

When evaluating FS for the five types of spectrum tenancy related measurement data in Table I, we randomly choose ten sequences each with 10^5 consecutive samples from the measurement data. We show the average of the FSs of the five features in the ten sequences in Fig. 2. We choose the tenancy of the 25th RB as the prediction target, and evaluate the FS of the five features within one second of the tenancy. Because there are 50 tenancy and MCS values each millisecond, we only show the FS of the MCS and the tenancy of the 25th RB due to space limitations in Fig. 2.

In Fig. 2, the FS values of the five prediction feature candidates, tenancy, MCS, the number of OFDM symbols, the number of DCIs, and the assignment types are shown in black, red, blue, green, and orange with different line and marker types as indicated by the legend. One observation on the results is that MCS, the number of OFDM symbols, and the number of DCIs achieve comparable FS values with the spectrum tenancy which has been the only prediction feature in existing studies. Thus, we will explore the performance of the three new features MCS, the number of OFDM symbols, and the number of DCIs in later parts of the paper. The other finding is that the FS values of the top four features decrease fast within 40 ms, so we adopt the four metrics within 40 ms of

the prediction target as the prediction features. Compared with other prediction works that use tenancy in several previous seconds [22], [29], the feature selection greatly reduces MDP state space size and time to accumulate training data. This finding also highlights the importance of spectrum usage data with high resolution on the prediction performance, because measurement results with low time granularity may fail to capture the most predictive features in 40 ms.

III. PROBLEM FORMULATION

We start from modeling spectrum prediction as an MDP that is capable of online parameter updates. The state space, the action space, the transition probabilities, and the reward function are explained as follows.

State space. Suppose that a PU system schedules transmissions on a set $\mathcal{C} = \{c_i | i \in \{1, \dots, N\}\}$ of N orthogonal channels in discrete time intervals $\mathcal{T} = \{t | t \in \mathbb{N}^+\}$. A prediction agent forecasts channel usage of the PU system according to observations $\mathbf{s}_t$ on some measurable physical quantities of the N RBs before time slot t , where states in all time instances fall in the space $\mathcal{S}$. We consider that the elements contained in the matrix $\mathbf{S}_t$ of random variables are flexible, as we will use several new features besides the binary tenancy, which will be further explained later in equations (12), (14), (15). Traditionally, most spectrum tenancy algorithms have only considered $\mathbf{S}_t$ as channel occupancy or signal strength in previous time slots [13], [22], [30].

Action space. The prediction results, i.e., actions taken by the agent, belong to the set $\mathcal{A} = \{A^i | A^i \in \{0, 1\}, i \in \{1, \dots, N\}\}$, where A^i is a random variable representing the prediction result for a single RB. If RB i is predicted to be idle at time t , then $A_t^i = 0$; otherwise, $A_t^i = 1$.

Transition probabilities. We assume that the evolution of states from one time slot to the next obeys the Markov property, meaning that $\mathbb{P}(\mathbf{s}_{t+1} | \mathbf{s}_t, a_t, \dots, \mathbf{s}_1, a_1) = \mathbb{P}(\mathbf{s}_{t+1} | \mathbf{s}_t, a_t)$, where $\mathbb{P}(\cdot)$ denotes the probability, and $a, \mathbf{s}$ are values assumed by random actions and states. The actions are governed by the prediction policy π that defines the probability of an action a in a state $\mathbf{s}$, i.e., $\pi(a | \mathbf{s}) = \mathbb{P}(A_t = a | \mathbf{s}_t = \mathbf{s})$.

Reward function. The tenancy prediction of each RB, A_t^i , obtains the rewards, $R_t^i \in \{1, -\tau_n, -\tau_p\}$, with three possible values in the next time slot, $t+1$, when the random variable of actual tenancy, $Z_{t+1}^i \in \{0, 1\}$, is observed. Similar to A_t^i , $Z_t^i = 0$ if channel c_i is idle; otherwise, $Z_t^i = 1$. The reward at time t for channel c_i is characterized by the random variable R_t^i , taking the three possible values defined as below,

$$R_t^i = \begin{cases} 1, & \text{if } Z_{t+1}^i = A_t^i, \\ -\tau_n, & \text{if } Z_{t+1}^i = 1, A_t^i = 0, \\ -\tau_p, & \text{if } Z_{t+1}^i = 0, A_t^i = 1. \end{cases} \quad (2)$$

When the predicted channel occupancy is the same with the actual tenancy in the next time slot, i.e., $Z_{t+1}^i = A_t^i$, the reward is 1. In FN cases, $Z_{t+1}^i = 1, A_t^i = 0$, the reward is $-\tau_n$; while in False Positive (FP) cases, the reward is $-\tau_p$.

The two parameters τ_n and τ_p take values in $[0, 1]$, depending on how strong the punishments are in the two cases.

The cumulative discounted reward, G_t , is a random variable denoting total rewards achieved after interval t , $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$. Rewards gained far from now are discounted by rate γ , $0 \leq \gamma \leq 1$, due to increased uncertainties and waiting. Define the state-action function $q_\pi(s, a)$ as

$$q_\pi(s, a) = \mathbb{E}_\pi(G_t | \mathbf{S}_t = s, A_t = a), \quad (3)$$

which is the expected rewards achieved by the agent when taking the action a in the state s under the policy π , and $\mathbb{E}(\cdot)$ is the expectation operator. Now that we have defined states, actions, transition probabilities, and rewards in the canonical MDP framework, the goal of the agent is to discover the optimal policy that maximizes the state-action values, i.e., $q_*(s, a) = \max_\pi q_\pi(s, a)$.

IV. THE BASELINE DQN ALGORITHM

Here we solve the prediction problem with the DQN algorithm that has achieved impressive results when applied to MDPs with similarly large state space sizes [31]. The DQN algorithm has received many improvements, and several major ones are combined and named Rainbow [25], which is adopted as our baseline solution to be upgraded by later sections.

Finding the optimal policy π^* requires full knowledge of how the environment changes, which is $\mathbb{P}(s_{t+1} | s_t, a_t)$ for actions in all states. However, estimating and storing these probabilities are impractical due to high computation costs and the stringent timeliness of spectrum predictions. In order to achieve online optimization of the state-action values without complete environment information, Q-learning is adopted to discover the optimal state-action values directly from data samples online [32]. In Q-learning, state-action values, or Q-values, are shifted towards optimality iteratively as below,

$$q(s_t, a_t) \leftarrow q(s_t, a_t) + \beta \left[r_{t+1} + \gamma \max_a q(s_{t+1}, a) - q(s_t, a_t) \right], \quad (4)$$

where $\beta \in [0, 1]$ is the learning rate. The prediction agent moves the state-action value of a new sample towards the best foreseeable based on the current knowledge of later rewards. The agent starts learning and updating Q-values upon the availability of data samples, without the need to accumulate large amounts of data for transition probability estimations.

The Q-learning agent stores state-action values for both the policy being updated, denoted as $\hat{q}$, and those of the target policy $\bar{q}$. In our experiments, we adopt the widely used ε -greedy policy to generate the training data. Therefore, the agent chooses the action with the highest expected reward with the probability $1 - \varepsilon$, and a random action with a small probability ε . Thus, the agent explores all state-action values, and the convergence to q_* is guaranteed.

Though Q-learning enables timely updates of state-action values without complete knowledge of the environment, the prediction agent still needs to gather innumerable samples of

Q-values (4). To avoid accumulating the large amount of Q-values, we approximate them $q_*(s, a)$ with a Convolutional Neural Network (CNN) $\hat{q}(s, a, \mathbf{w})$ with weights $\mathbf{w}$, to generalize from the limited amount of observed state-action values to unknown ones. To achieve convergence of the parameters in the DQN, many techniques are proposed in [25], [31], we briefly explain them here for self-containment purposes.

To combat the instability and divergence caused by non-linear approximators, such as the CNN, the encountered transitions, $e_t = (s_t, a_t, r_t, s_{t+1})$ are stored in an experience replay buffer $\mathcal{H}$. At each time step, the prediction agent fetches a batch of K experiences from the buffer to learn how to update the weights. Each sample e_t is associated with a priority p_t that determines the probability of fetching e_t , $P_t = \mathbb{P}(\text{take the sample } e_t) = p_t^\alpha / \sum_i p_i^\alpha$, where α is a parameter that determines the priority of p_i . The current Prioritized Experience Replay (PER) scheme assigns the priority based on TD errors, i.e.,

$$p_t = |\delta_t| = |r_{t+1} + \gamma \max_a \bar{q}(s_{t+1}, a_{t+1}, \bar{\mathbf{w}}_t) - \hat{q}(s_t, a_t, \hat{\mathbf{w}}_t)|. \quad (5)$$

As pointed out in [33], the prioritized replay introduces a bias towards the estimated values, because the distribution of e_t which affects the weights in the DQN is different from that in the natural appearance of e_t . To counterbalance the bias, we adopt the solution of importance-sampling weights, $w_i = p_i^{-\rho}$ which is applied to Q-learning update. When the parameter $\rho = 1$, the effects of sampling the transitions with different priorities are fully compensated by weight updates.

Another improvement to the original DQN is the dueling network structure that separates state-action values into values and advantages as below.

$$q(s, a, \mathbf{w}) = v(f(s, \mathbf{w}_1), \mathbf{w}_2) + a(f(s, \mathbf{w}_1), a, \mathbf{w}_3) - \frac{\sum_{a'} a'(f(s, \mathbf{w}_1), a', \mathbf{w}_3)}{|\mathcal{A}|}, \quad (6)$$

where $|\cdot|$ is the size of a set, $v(\cdot)$ and $a(\cdot)$ are the value function and the advantage function. To propagate newly observed rewards to former states faster, we adopt multi-step bootstrapping [32]. Thus, the TD error δ_t becomes

$$\delta_t = r_{t+1}^{(n)} + \gamma_t^{(n)} \max_a \bar{q}(s_{t+n}, a_{t+1}, \bar{\mathbf{w}}_t) - \hat{q}(s_t, a_t, \hat{\mathbf{w}}_t), \quad (7)$$

where $r_t^{(n)} = \sum_{k=0}^{n-1} \gamma_t^{(k)} r_{t+k+1}$, and $\gamma_t^{(n)} = \prod_{k=0}^{n-1} \gamma_{t+k+1}$.

Another upgrade is to find out the distribution of the Q-values. Assume that the support for the distributions is a vector $\mathbf{z} = \{z_1, \dots, z_M\}$, and M is the number of elements. Hence, the state-action value $q(s_t, a_t, \mathbf{w}_t)$ is now determined by the distribution $d_t = (\mathbf{z}, \mathbf{p}(s_t, a_t, \mathbf{w}_t))$. Since the optimization target is now the distribution of state-action values on the support, the TD error is defined as the Kullback-Leibler divergence between d_t and $d_t^{(n)}$,

$$d_t^{(n)} = \left(r_{t+1}^{(n)} + \gamma_t^{(n)} \mathbf{z}, \bar{\mathbf{p}}(s_{t+n}, a_{t+n}^*, \bar{\mathbf{w}}_t) \right), \quad (8)$$

where $a_{t+n}^* = \arg \min_{a_{t+n}} \pi(a_{t+n} | s_{t+n})$. We denote the divergence as $D_{KL}(\Phi_{\mathbf{z}} d_t^{(n)} || d_t)$, where $\Phi_{\mathbf{z}}$ is the L2 projection

of the target distribution onto the support $\mathbf{z}$. The number of the outputs in the last advantage layer is $|\mathcal{A}| \times M$, and the softmax function obtains the distribution of the rewards.

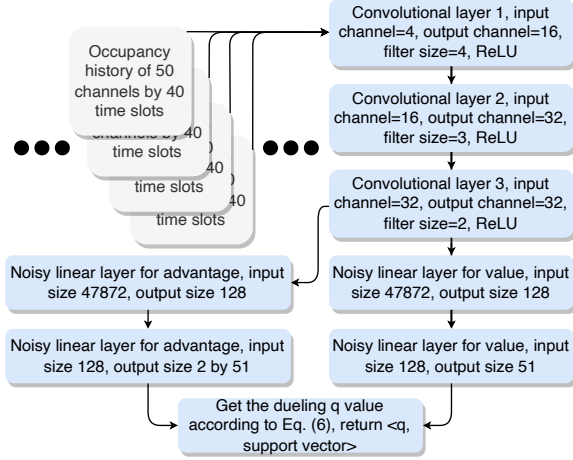


Fig. 3: The structure of the CNN.

The above Rainbow DQN [25], is adopted as the baseline and summarized in Algorithm 1. The agent initializes CNN weights, and then starts observing and predicting the spectrum. Each step, a batch of experiences are sampled according to their priorities based on TD errors. Then, the agent computes the weight updates, and the TD errors in the online network. Every C steps, the weights of the target are updated by cloning the online network. Lastly, the agent conducts the ϵ -greedy prediction of the spectrum tenancy.

The structure of the CNN network is illustrated in Fig. 3. Four matrices of spectrum tenancy go through three convolutional layers with Rectified Linear Units (ReLUs) as activation functions. The outputs are then processed by two streams of noisy linear blocks [34], to obtain the values and advantages in the dueling DQN. Lastly, we sum the value and the advantage streams according to (6) to obtain the Q-value distribution, and the Q-value is calculated as the inner product of the distribution and the support vector.

For the baseline solution, we consider the spectrum tenancy in 40 previous milliseconds and 50 RBs as the state space as explained in Sec. II-B. Thus,

$$\mathbf{S}_t = \begin{pmatrix} Z_{t-40}^1 & \cdots & Z_{t-1}^1 \\ \vdots & \ddots & \vdots \\ Z_{t-40}^{50} & \cdots & Z_{t-1}^{50} \end{pmatrix}. \quad (9)$$

To predict the spectrum tenancy a_t , the four channels to the first convolutional layer are $\mathbf{S}_{t-3}, \dots, \mathbf{S}_t$. For the reward parameters, we set $\tau_n = 1, \tau_p = 0$ because FNs are more harmful due to potential collisions. Among the 50 downlink channels, we predict the tenancy of the 25th.

The four key performance metrics of the baseline scheme are illustrated in blue in Fig. 4. The loss is the TD error, and the loss at every time step is shown up to 2×10^4 milliseconds. To obtain the prediction accuracy, the FP rates, and the FN rates, we cut the 2×10^4 running steps into segments of 100

Algorithm 1 The baseline Rainbow DQN algorithm

Input: Data sample e_t , learning rate β , priority strength α , explore factor ϵ , weight update step C , batch size K .

Output: Predictions a_t that maximizes $\bar{q}(\mathbf{s}_t, a_t, \bar{\mathbf{w}}_t)$

- 1: Initialize replay buffer $\mathcal{H} = \emptyset$, $\Delta = 0$
- 2: Initialize weights of $\bar{q}$, $\hat{q}$ to be random values
- 3: **for** $t = 1, 2, \dots$ **do**
- 4: Store $e_{t-1} = (\mathbf{s}_t, a_t, r_t, \mathbf{s}_{t+1})$ in $\mathcal{H}$ with maximal priority $p_{t-1} = \max_{i < t-1} p_i$
- 5: **for** $i = 1, \dots, K$ **do**
- 6: Sample transition i with probability $P(i)$
- 7: Calculate importance-sampling weight w_i
- 8: Compute the multi-step Kullback-leibler divergence based difference $D_{KL}(\Phi_{\mathbf{z}} d_i^{(n)} || d_i)$ according to (8)
- 9: Update transition priority P_i
- 10: Accumulate weight $\Delta \leftarrow \Delta + w_i D_{KL} \nabla \hat{q}(\mathbf{s}_i, a_i, \hat{\mathbf{w}}_t)$
- 11: **end for**
- 12: Update weight $\hat{\mathbf{w}} \leftarrow \hat{\mathbf{w}} + \beta \Delta$, and reset $\Delta = 0$
- 13: **if** $0 == \text{mod}(t, C)$ **then**
- 14: Update weights of target network, $\bar{\mathbf{w}} \leftarrow \hat{\mathbf{w}}$
- 15: **end if**
- 16: With probability ϵ , greedily choose action $a_t \leftarrow \max_{a_t} \bar{q}(\mathbf{s}_t, a_t, \bar{\mathbf{w}}_t)$; otherwise, randomly choose a_t
- 17: **end for**

steps in which the ratio of the number of correct predictions, FPs, and FNs over 100 are computed as the accuracy, FP, and FN. The average accuracy, FP rate, and FN rate in the 2×10^4 time steps are 0.8858, 0.0674, and 0.0468. All the metrics are evaluated on test data in this paper, meaning that the agent has not encountered the experiences before making the predictions. Though the baseline DQN is able to ramp up the prediction accuracy given limited training samples, there is still room for improvements, as there are sudden accuracy drops below 0.5 and the overall accuracy 0.8858. To further improve the performance, two schemes will be introduced in the next two sections.

V. THE TENANCY DISTANCE BASED EXPERIENCE REPLAY

In this section, we improve the accuracy by proposing a new experience replay scheme for the DQN algorithm.

As pointed out in [25], the PER scheme is the most influential performance booster among the proposed DQN updates. By giving priorities to the experiences causing larger TD errors when sampling from the replay buffer, the agent learns the weights to lower the losses sooner. Motivated by this scheme, we redefine the priority to include tenancy distances between the current state and those in the replay buffer.

We define the tenancy distance between two states as the Frobenius norm of their difference. Thus, the tenancy distance between states $\mathbf{s}_i$ and $\mathbf{s}_j$ is

$$\text{dist}(\mathbf{s}_i, \mathbf{s}_j) = \|\mathbf{s}_i - \mathbf{s}_j\|_F = \sqrt{\sum_{k=1}^{40} \sum_{l=1}^{50} (s_{i,kl} - s_{j,kl})^2}, \quad (10)$$

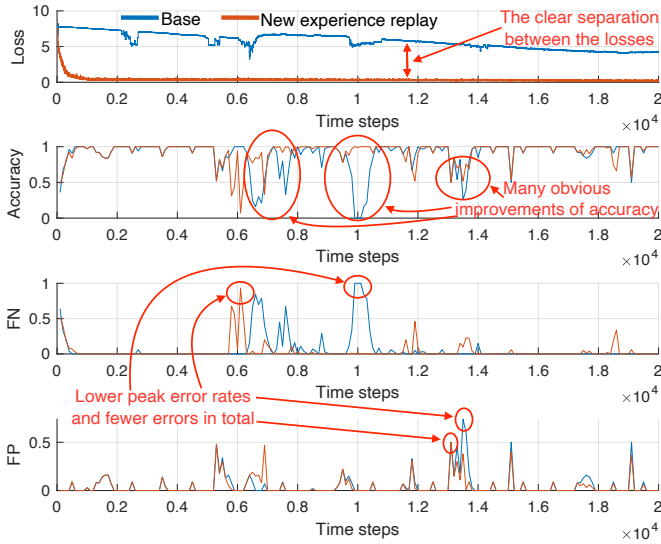


Fig. 4: Performance comparisons between the baseline DQN and the one with the upgraded PER scheme.

where $\|\cdot\|_F$ denotes the Frobenius norm, and $s_{i,kl}, s_{j,kl}$ are the elements in the matrices s_i and s_j . At time step t , the priority of experience i in the batch with K experiences are

$$p_i = \frac{K|\delta_i|}{\sum_j |\delta_j|} + \frac{\sum_j 1/\|s_j, s_{t+1}\|_F}{K\|s_t, s_{t+1}\|_F}, \quad (11)$$

where the TD error and the tenancy distance are normalized by their corresponding arithmetic average. The normalization ensures that TD errors and tenancy distances exert similar influence on the combined priorities.

The four metrics of the new PER are depicted in orange color in Fig. 4, while the performance of the baseline is in blue. We observe that the DQN with the new PER not only achieves much smaller losses, but the losses exhibit fewer fluctuations. Though small and steady losses may indicate overfitting, the improved accuracy proves that the new algorithm performs better than the baseline. The new algorithm also achieves fewer errors, and lowers peak error rates as marked in the figure. The accuracy, FP, and FN rates are 0.9259, 0.0392, and 0.0348, respectively. Thus, the new scheme brings the prediction error rates down by 35.11%, significantly reducing the prediction errors.

More importantly, the enhanced DQN with the new prioritized experience replay achieves much sooner convergence than the baseline, as shown in Fig. 4(a). By prioritizing the buffered experiences that exhibit large TD errors and small sample distance to current situations, we achieve stable loss in around 2000 time steps, i.e. 20 seconds, while the loss of the baseline becomes stable after encountering more than 18000 samples. Thus, our new DQN achieves a substantially reduced sample complexity that is only 11.1% of the baseline.

VI. THE PREDICTION BASED ON VARIOUS FEATURES

Besides the improved PER, the other performance booster is the new features introduced in Section II-B. In this section, we for the first time improve the spectrum tenancy prediction by introducing new inputs obtained by decoding LTE DCIs

[21]. We consider three data fields, the MCS, the number of OFDM symbols, and the number of DCIs.

To accommodate MCS values as prediction features, the states of the MDP are redefined as

$$\mathbf{s}_t^M = \begin{pmatrix} Z_{t-40}^{M,1} & \cdots & Z_{t-1}^{M,1} \\ \vdots & \ddots & \vdots \\ Z_{t-40}^{M,50} & \cdots & Z_{t-1}^{M,50} \end{pmatrix}, \quad (12)$$

where $Z_t^{M,i} \in \{0, \dots, 32\}$. The reward function is revised accordingly as follows,

$$R_t^i = \begin{cases} 1, & \text{if } Z_{t+1}^i = A_t^i, \text{ or } Z_{t+1}^i > 1, A_t^i = 1, \\ -\tau_n, & \text{if } Z_{t+1}^i \geq 1, A_t^i = 0, \\ -\tau_p, & \text{if } Z_{t+1}^i = 0, A_t^i = 1, \end{cases} \quad (13)$$

since $Z_t^{M,i}$ utilizes the MCS value plus one to represent the channel being occupied. Different from MCS values that are scheduled for each LTE RB, the number of OFDM symbols and the number of DCIs reflect the spectrum usage decisions made every millisecond for all the RBs. To incorporate the number of OFDM symbols of the control channel, the state of the MDP is redefined as

$$\mathbf{s}_t^S = \begin{pmatrix} Z_{t-40}^1 & \cdots & Z_{t-1}^1 \\ \vdots & \ddots & \vdots \\ Z_{t-40}^{50} & \cdots & Z_{t-1}^{50} \\ Z_{t-40}^S & \cdots & Z_{t-1}^S \end{pmatrix}, \quad (14)$$

where the last row in the above matrix is the number of OFDM symbols of the control channel, and $Z_t^S \in \{1, 2, 3\}$ since the LTE control channel can occupy the first one, two, or three OFDM symbols every millisecond. The spectrum resources within the downlink control channel carry a number, Z_t^N , of DCIs which include information such as the MCS. To accommodate the number of DCIs in the MDP state, we revise the state definition as

$$\mathbf{s}_t^N = \begin{pmatrix} Z_{t-40}^1 & \cdots & Z_{t-1}^1 \\ \vdots & \ddots & \vdots \\ Z_{t-40}^{50} & \cdots & Z_{t-1}^{50} \\ Z_{t-40}^N & \cdots & Z_{t-1}^N \end{pmatrix}, \quad (15)$$

where Z_t^N is a natural number, $Z_t^N \in \mathbb{N}$.

As the new prediction features enlarge the size of the state space, it is possible that the prediction agent would require high sample complexity, the amount of samples for the prediction agent to learn the optimal weights of the network. Thus, we examine the analytic upper bounds of the sample complexity caused by newly introduced prediction features before evaluating their performance through simulations.

A. Impacts of additional features on sample complexity

Since the MDP state with MCS $\mathbf{s}_t^M$ has the largest space size among the three new features due to the addition of 31 possible values to each element $Z_t^{M,i}$ in the matrix, we

evaluate its sample complexity under two criteria. Under the classical criterion to approximate optimal Q-values, we adopt the results proposed in [35]. Besides the classical sample complexity bound, we propose another new upper bound of sample complexity required by the agent to stably predict spectrum tenancy.

1) *Estimating the classical sample complexity of the DQN with MCS as prediction inputs:* Existing studies on the sample complexity of RL are based on the stopping criterion that the iterative algorithm has learned the Q-values which are close enough to the optimal, meaning that

$$|q_l(s, a) - q_*(s, a)| < \xi, \forall s, a, \quad (16)$$

where $q_l(s, a)$ is the estimated Q-value of taking action a in the state s after having learned l samples, and ξ is a small positive number indicating the tolerable difference to deem the convergence of Q-values. According to [35], the amount of samples required by an RL learner to reach the classical stopping criterion is upper bounded by

$$\frac{|S||\mathcal{A}|T}{\min P(s)} (\log(|S|/\delta) + \log(\log(1/\xi))), \quad (17)$$

where $|\cdot|$ denotes the cardinality of a set, T is the mixing time of the underlying Markov chain, $P(s)$ is the steady state probability of state s , and δ is a small positive number such that $1 - \delta$ is the probability that (16) is satisfied.

To gain an insight into the sample complexity of the two MDPs with different prediction features under the classical stopping condition, we estimate the values of all the parameters in (17) from the measurement data described previously, and obtain the upper bound of sample complexity. Based on 6000 samples, we estimate the variables and the upper bounds of the sample complexity of two MDPs with different states, and the results are shown in Table II. Since the occupancy of only one RB is predicted, we have $|\mathcal{A}| = 2$ for both MDPs. For the MDP using history tenancy as its state, $|S| = 1665$ different states are observed in the 6000 samples. The mixing time, the number of steps for the total variation distance to decay by a factor of e , is 255.8, and the minimum probability of a state in the steady state is 1.721×10^{-4} . Thus, the sample complexity is 5.225×10^{10} according to (17). The MDP using MCS as its state has a much larger state space of the size of 4145, shorter mixing time of 58.3, and a slightly smaller $\min P(s_0) = 1.66 \times 10^{-4}$. As a result, introducing the MCS as new features does not increase the classical sample complexity required to approach the optimal Q-values.

TABLE II: Classical analytic upper bounds of sample complexity of the MDPs with different states

States	Bounds	$ S $	$ \mathcal{A} $	T	$\min P(s_0)$
Tenancy	5.225×10^{10}	1665	2	255.8	1.721×10^{-4}
MCS	3.338×10^{10}	4145	2	58.3	1.66×10^{-4}

2) *The sample complexity based on prediction stability:* Though the sample complexity satisfying the condition given in (17) sufficiently guarantees that an agent makes decisions based on Q-values close to the optimal, judging convergence in this way may require more training samples than necessary.

This is due to the fact that the behavior of an ε -greedy Q-learning agent depends only on the values of taking different actions in a state. When the agent is in the exploration state with probability ε , Q-values are not considered; the action generating the largest Q-value will be taken when the agent is greedy with probability $1 - \varepsilon$. Thus, we propose that a Q-learning agent predicts stably when the actions producing the largest Q-values are no longer likely to change. Because there are only two possible actions in every state, the number of samples required to achieve stable prediction for state s is

$$k = \arg \min_k (\inf_{m \geq 0} (\mathbb{P}\{q_{k+m}(s, 1) - q_{k+m}(s, 0) \geq 0\}) \geq 1 - \epsilon_0), \quad (18)$$

where $q_k(s, a)$ is the Q-value learned by the agent who has come across the samples of the state-action pair for k times, and ϵ_0 is a small positive value. The definition in (18) exists under the assumption that the probability of the predicted RB being occupied, denoted as a , is no less than the complementary case, i.e. $a \geq 0.5$, for a state s . When $a < 0.5$, the definition and the solution of the sample complexity k can be obtained by similar means, so they are omitted for brevity.

According to (4), $q(s, a)$ changes by $\beta[r_{t+1} + \gamma \max_a q(s_{t+1}, a) - q(s_t, a_t)]$ after learning a sample. Since we focus on the beginning stages when Q-values are close to the initial value of 0, the updates are dominated by the reward terms. Because the rewards are all multiplied by the constant β , we can ignore it for the purpose of studying the required number of samples k to reach the separation condition in (18). We also assume that the exploration parameter ε is small enough to be ignored to derive the distribution of distance changes caused by each sample. Thus, the separation between the Q-values of the binary actions in a state after encountering the samples n times, $\Delta d_n(s)$, can be approximated by a normal random variable,

$$\Delta d_n(s) \sim N(n((1 + \tau_p)a - \tau_p), n(a(1 - a)(2 - a)(1 + \tau_p)^2)), \quad (19)$$

according to central limit theorem. Substituting the separation in (18) with $\Delta d_n(s)$, we get

$$k = \arg \min_k (\inf_{m \geq 0} (\mathbb{P}\{\Delta d_{k+m}(s) \geq 0\}) \geq 1 - \epsilon_0). \quad (20)$$

Because $\Delta d_n(s)$ is distributed as (19), it can be shown that $\mathbb{P}\{\Delta d_n(s) \geq 0\}$ is monotonously increasing with regard to n . Hence, (20) is further simplified to

$$k = \arg \min_k (\mathbb{P}\{\Delta d_k(s) \geq 0\} \geq 1 - \epsilon_0). \quad (21)$$

Solving (21), we get

$$k = \left\lceil \left(-\Phi^{-1}(\epsilon_0) \sqrt{\frac{(1 - a)(2 - a)}{4a}} + \sqrt{\frac{(-\Phi^{-1}(\epsilon_0))^2(1 - a)(2 - a)}{4a} + \frac{\tau_p}{a(1 + \tau_p)}} \right)^2 \right\rceil, \quad (22)$$

where $\Phi^{-1}()$ is the inverse cumulative distribution function of the standard normal random variable. For a in range $[0.5, 1]$, k

is a monotonically decreasing function of a . Thus, the larger the a value, the fewer samples are required to reach stable prediction with the probability of at least $1 - \epsilon_0$.

According to the above analysis, the stable prediction in a state s_0 requires k_{s_0} samples in that particular state. Thus, the expectation of the total number of samples to guarantee the existence of k_{s_0} samples in the state s_0 from an arbitrary sample path of an MDP with stationary distribution $P(s)$ is $k_{s_0}T/P(s_0)$, where T is the mixing time of the MDP. Hence, the upper bound of sample complexity for stable prediction is

$$\max_{s_0} k_{s_0}T/P(s_0). \quad (23)$$

Compared with the classical upper bound in (17) which depends on the size of the state space $|S|$ and the size of the action space $|A|$, the upper bound of sample complexity under the new stopping criterion is much smaller as shown in Table III. To obtain k , we let $a = 0.5$, $\epsilon_0 = 0.001$, and $\tau_p = 0$ as discussed previously to suppress FNs. We observe that the MDP with MCS in its states has slightly smaller sample complexity to achieve stable prediction results.

Based on the results in Table II and Table III, we conclude that adding the new features does not cause the DQN agent to learn extra samples to approximate the optimal Q-values, or to achieve stable prediction results.

TABLE III: New analytic upper bounds of sample complexity of the MDPs with different states

States	Bounds	k	T	$\min P(s_0)$
Tenancy	2.23×10^7	15	255.8	1.721×10^{-4}
MCS	5.268×10^6	15	58.3	1.66×10^{-4}

B. Performance evaluations for DQNs with new features

Since the DQNs with enlarged state spaces do not have greater sample complexity, we evaluate the performance of the three DQNs until the same 2×10^4 time steps with the baseline case. Most other parameters are also the same with the base case in Section IV, except that some parameters in the CNN network need to be updated to accommodate the extra row of features introduced by the number of OFDM symbols and the number of DCIs as shown in (14) and (15).

TABLE IV: Average performance metrics of the five DQNs

Schemes	Base	Replay	MCS	DCIsize	NDCI
Accuracy	0.8858	0.9259	0.9286	0.9269	0.9367
FN	0.0674	0.0348	0.0206	0.0224	0.0126
FP	0.0468	0.0393	0.0508	0.0507	0.0507

The average prediction accuracy, FNs, and FPs of the five prediction schemes are listed in Table IV. We observe that the three prediction schemes with additional spectrum tenancy information decoded from LTE downlink control channel achieve much better results than the baseline, due to the accuracy increase about 5.5%, and the decrease in FNs around 70%. Among the three new features, the number of DCIs exerts the most beneficial impacts on prediction performance, highlighted by the best accuracy of 0.9367, and the lowest FN of 0.0126 among the five schemes.

Since the three new features obtain similar prediction performance and the number of DCIs is the best among them, we illustrate its performance in the first 2×10^4 running steps

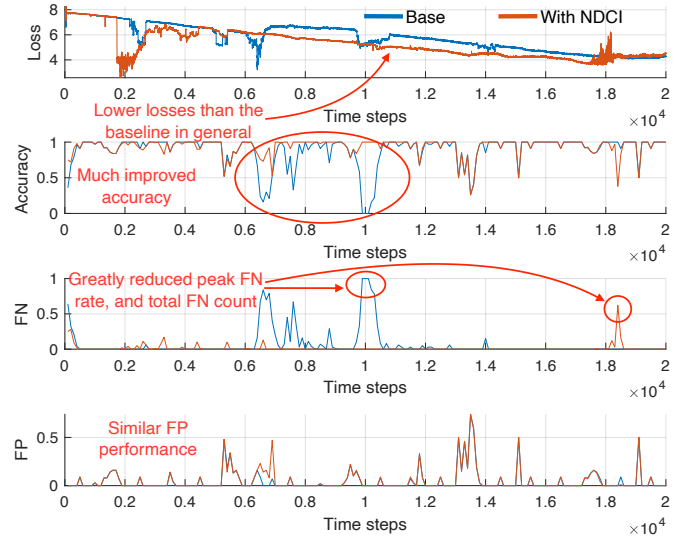


Fig. 5: Performance comparisons between the baseline DQN and the one with the number of DCIs.

along with the baseline performance achieved by the state-of-the-art DQN algorithm in Fig. 5. The improvement in loss is not as good as that of the updated PER scheme shown in Fig. 4, but the new feature still reduces loss fluctuations and lowers the loss. In terms of prediction accuracy, several sudden deep accuracy drops in the base scheme are ameliorated by the addition of new features. The decrease of FNs induced by the new feature can be clearly spotted, in the aspects of both peak FN rate and the total occurrence of FNs. Unlike in Fig. 4 where the FP is reduced by the new PER algorithm, the two FP rates in Fig. 5 are similar. Thus, the DQN with new features is slightly better than the upgraded PER scheme, because FNs may give rise to collisions between SUs and PUs while FPs only decrease the amount of spotted spectrum holes.

VII. PERFORMANCE EVALUATIONS AND COMPARISONS

In this section, we conduct detailed evaluations under various spectrum usage levels, and compare with one of the most accurate non-DQN predictions.

A. Predict spectrum tenancy with different occupancy levels

The performance evaluations in previous sections are conducted by applying the algorithms to a time period when the spectrum usage is around 50%, meaning that there is an equal amount of occupied and unoccupied spectrum slices. Since it is well known that imbalanced data may train defective predictors [36], now we evaluate the performance of the proposed algorithms in such adverse situations.

TABLE V: Performance under biased spectrum usage.

Usage	Schemes	Base	Replay	NDCI
Around 7%	Accuracy	0.7743	0.8226	0.8413
	FN	0.1995	0.1562	0.1239
	FP	0.0262	0.0212	0.0348
Over 90%	Accuracy	0.9392	0.9464	0.9506
	FN	0.0239	0.0168	0.0117
	FP	0.0369	0.0368	0.0377

We validate the performance of the algorithms when applied to two extreme scenarios where the overall spectrum usage is as low as 7% and as high as above 90%, and the results are recorded in Table V. As shown in the table, our algorithms outperform the baseline in both cases.

TABLE VI: Prediction performance for different RBs.

RB	Schemes	Base	Replay	NDCI
# 1	Accuracy	0.8843	0.9264	0.9119
	FN	0.0663	0.0244	0.0318
	FP	0.0494	0.0492	0.0563
# 50	Accuracy	0.8897	0.8921	0.8957
	FN	0.0432	0.0404	0.0368
	FP	0.0671	0.0675	0.0675

B. Predict spectrum tenancy in different RBs

We apply our algorithms to predict the spectrum tenancy of all the 50 RBs. Due to space limitations and the similarities among the results of different RBs, we present the outcome of predicting the two RBs on the ends of the spectrum band, RB 1 and RB 50. The results are presented in Table VI. The DQN with the new PER and the new prediction feature of the number of DCIs perform better than the state-of-the-art DQN.

C. Performance comparison with a non-DQN algorithm

Among the large variety of spectrum tenancy prediction algorithms, the accuracy of MLP is at the top, since the predicted energy levels have a root mean square error as small as 0.04 from the actual energy levels normalized to one [26]. Hence, we apply the MLP algorithm to our data for performance comparisons.

We adopt a feed-forward MLP with three layers as suggested in [26]. Different from the 4-5-1 network structure that has four neurons in the input layer, five in the hidden layer, and one in the output layer, we use the 6-10-1 structure for the MLP model after experimenting with different structures. The activation function is the sigmoid, and the inputs to the MLP algorithm are the binary spectrum tenancy in six previous time slots in the RB of interest. Six different training data sizes are evaluated, ranging from 200 to 2000 as shown by the horizontal axis in Fig. 6. When enough training iterations have been conducted on a training data set and the accuracy stabilizes, the trained MLP is evaluated on a test data set. The accuracy versus training data volume is illustrated in Fig. 6.

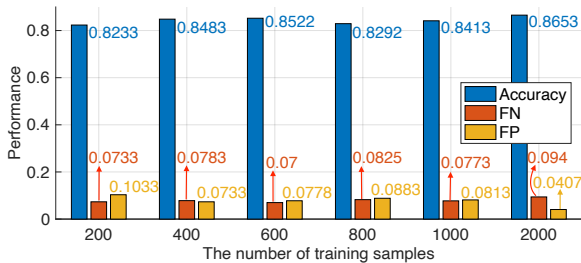


Fig. 6: MLP accuracy versus training data volume .

According to Fig. 6, the prediction accuracy of MLP increases slowly as the training data size grows. When the MLP model is trained with a data set of 1000 samples, the accuracy is around 84%. In comparison, our DQNs with the upgraded PER and new features achieve stable prediction accuracy of

around 92% after encountering 1000 samples, as shown in Fig. 4 and 5. When the training data size reaches 2000 samples, the MLP boosts its accuracy to around 86%. In comparison, our proposed DQN shows an increase in prediction accuracy up to 14%, from around 82% to over 93%. The MLP has similar FNs and FPs, since it cannot suppress FNs as the DQNs do by adjusting the reward parameters.

To sum up, our proposed algorithms achieve more accurate prediction results after learning fewer samples than the MLP which does not have runtime model update capabilities.

VIII. RELATED WORK

Data exerts fundamental influence on prediction performance, but existing spectrum predictions have unanimously relied on low resolution spectrum data collected by the classical energy detection method [22], [26], [37], [38]. For instance, a spectrum prediction system targeting GSM bands [26], the training and validation data sets are measured at very poor time granularity of 157 seconds. The data is further averaged before feeding into the prediction agent, resulting in only 548 tenancy samples in 24 hours, which is way slower than how frequently GSM spectrum resources are scheduled. Though new sensing and measurement methods enabling finer granularities have been proposed in recent years [24], [39], low resolution data are still the mainstay for spectrum prediction [40]. Besides low resolutions, errors are inevitable when converting PSD values to actual binary spectrum tenancy according to artificially chosen threshold [20]. As high-resolution spectrum tenancy changes fast, it is important that the prediction algorithms are capable of timely runtime model updates. However, since existing predictions target low resolution data often assumed to have stable dynamics, most prediction algorithms do not consider online update capabilities of model parameters [41], [42]. To remedy the low resolution data and the ensuing predictions algorithms with little consideration on runtime update capabilities, we propose the improved DQN schemes that are capable of accurately inferring spectrum tenancy in fine granularities with new prediction features under reasonable sample complexities.

IX. CONCLUSION

Spectrum tenancy prediction is of utmost importance for wireless coexistence. Different from existing studies that are based on static prediction models and validated by data sets with low time-frequency resolutions, we propose accurate online algorithms based on DQNs whose performance is improved by the upgraded PER scheme and new prediction features. We analytically show that the newly discovered prediction features improve accuracy without increasing sample complexity. Thorough performance evaluations are conducted by applying the original and our improved DQNs to predict the binary spectrum tenancy of a commercial LTE cell. Our schemes significantly outperform the state-of-the-art DQNs and the MLP in the aspects of both sample efficiency and prediction accuracy.

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